A. Introduction/ Situation/ Background Information

In this assignment, you will apply a wide range of R programming techniques to analyze statistical data from a real-world dataset. The goal is to enhance your understanding of key statistical concepts while developing practical skills in R for data analysis. This assignment encourages exploration and application of statistical methods to derive meaningful insights from the data. It also improve the ability to communicate data-driven conclusion to stakeholder effectively.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 1 Apply a wide range of R operations to analyze statistical data.

D. Instruction to Students

This assignment is **an Group Assignment with 5-6 persons per group.** Students need to create 1 report. The students should submit the following to the Moodle:

1. **Document A:** i) Assignment Cover Page and ii) Report of the Task. Saved in StudentID.docx and StudentID.pdf format. For example: DSCxxx.docx and DSCxxx.pdf.
2. **Document B**: Export your source code into the format StudentID.R. For example: DSCxxx.R.
3. **Power Point C**: Create Power point slide for your report. Saved as StudentID.pptx. For example: DSCxxx.pptx. Every student has to present and will be evaluated accordingly.
4. The submission deadline of Assignment is on 23:55, 7 Dec 2026. Overdue penalty (20 marks deduction per day) will be given to project that submitted after deadline. Submission type: Softcopy
5. Submission type: **Softcopy**.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Task 1	100
	TOTAL	**100**

F. Task(s)

In this assignment, you will apply statistical techniques using R to investigate one of the three provided research topics using a real-world raw dataset.

i. Electric Vehicle Dataset: An Electrical Vehicle technology related dataset where the focus could be the Vehicle Specification, Performance, Population, or Charging Patterns.

ii. AI Tools Impact or Adoption Dataset: An Artificial Intelligence tools related dataset where the focus is to determine the impact or adoption in the works.

iii. WiFi Sensing for Human Activity Recognition: A signal processing related dataset where the focus is to analyse the signals to classify the huma action.

The data require careful exploration and feature extraction before meaningful analysis can be performed. You are expected to identify and derive relevant statistical features or indicators from the raw data, apply appropriate statistical analysis and modeling techniques, and provide the sumaarise of your finding.

Following are the Statistical methods studies in the classroom. You need to select the suitable techniques to use on the dataset, No Necessary apply All. In a case where the following technique is inapplicable due to the natural of dataset, you can apply other techniques.

Statistical method	List of Common Techniques (But not limited)
Descriptive Statistics	• Measuring Central Tendency: *Calculate and interpret the mean, median, and mode.* • Measuring Location via Standard Scores: *Calculate and interpret z-scores.* • Measuring Variability: *Calculate and interpret the variance and standard deviation.* • Covariance and Correlation:

	Analyze the relationship between variables using covariance and correlation (both Pearson and Spearman/Kendall as applicable). • Measuring Symmetry: *Analyze skewness and kurtosis of the dataset.*
Visualizations	• Frequency distributions and tables • Pie charts and bar charts • Boxplots • Histograms • Line graphs • Scatterplots **Saving Graphics**: Export each visualization as a PNG file.
Probability and Distribution Analysis	• Discrete Probability Distributions *Identify and model a discrete process if relevant.* • Relating Discrete Probability to Normal Probability: *Discuss and apply relevant probability concepts.* • The Sampling Distribution of Means: *Simulate and analyze the sampling distribution for one of your variables.*
Hypothesis Testing	• One-Sample Z-Test: *Conduct and interpret a one-sample z-test on a variable.* • One-Sample t-Test: *Perform and interpret a one-sample t-test.* • Paired-Samples t-Test: *Perform and interpret a paired-samples t-test.* • Two-Sample t-Test: *Compare means of two groups using a two-sample t-test.*
Regression Analysis	• Simple Regression: *Fit a simple regression analysis and discuss the findings.* • Multiple Regression Equation: *Fit a multiple regression model and discuss the findings.* • Logistic Regression: *Fit a logistic regression analysis and discuss the findings.* • Confidence and Prediction Intervals: *Compute confidence and prediction intervals for the regression model.*

1. Report structure

Your report must follow this research-oriented flow:

I. **Context and Data Understanding**

- Briefly describe the dataset and its context, including the type of data collected, observations, variables/features, and measurement characteristics.
- Explore the variability and patterns within the dataset and identify suitable statistical features or indicators that summarize the data effectively (e.g., mean, variance, standard deviation, trends, distributions, correlations, and other relevant domain-specific metrics).

II. **Data Pre-processing**

- Use appropriate tools or programming languages (e.g., R or Python) to clean and prepare the raw data for analysis. This may include identifying and handling missing values, removing duplicates or outliers, correcting inconsistencies, normalizing or standardizing variables, and filtering noisy records.

III. **Statistical Analysis and Modeling**

- Conduct suitable statistical tests, exploratory analysis, and/or predictive modeling techniques relevant to the dataset and research objectives.
- Clearly document the methods used, justify their selection, and present the results with appropriate interpretations, visualizations, and statistical evidence.

IV. **Actionable Conclusion and Insight**

- Summarize the key findings derived from the analysis and discuss their practical implications.
- Explain how the statistical insights or model outcomes can support decision-making, improve system performance, solve the identified problem, or contribute to future improvements and research directions.

2. Report format:

The report content should be organized with titles and subtitles.

- **Context**: Include all relevant visual outputs (charts/graphs) and your analytical findings (result). Regarding the R script, do not include the full code within the document; instead, simply reference the subsections (e.g.: The code of this finding could be found on sub-section xx in R code)
- **Titles and subtitles**:14 points font size, Times New Roman.
- **The content body**: 12 points font size, Times New Roman, 1.5 spacing and justify aligned.
- Spell and grammar checked.

3. Communicating the Insight from Dataset

Present your analysis, findings, and interpretations. You will be required to deliver using slides created with graphic presentation software, like Microsoft PowerPoint. The overall presentation will be 30 minutes (25 minutes for slide presentation and 5 minutes for Q&A session).

APPENDIX 1

MARKING RUBRICS

Component Title		Assignment				Percentage (%)	100	
Criteria		Score and Descriptors					Weight (%)	Marks
		Excellent (5)	Good (4)	Average (3)	Need Improvement (2)	Poor (1)		
	Data Pre-processing & Feature Engineering	Exceptional and innovative feature extraction (e.g., Phase Coherence). Robust and clearly justified cleaning of raw hardware errors.	Good feature extraction. Raw data cleaning is mostly effective with minor oversights.	Basic feature extraction. Minimal data cleaning applied.	Feature extraction is unclear or poorly executed. Cleaning is ignored.	No meaningful feature extraction or cleaning. Relies on uncleaned data.	**15**	
	Descriptive Stats & Correlation	Comprehensive analysis of central tendency, variability, and symmetry. Insightful interpretation of correlation.	Most statistics calculated correctly with good interpretation.	Appropriate statistics applied but basic interpretation.	Some statistics applied but poorly interpreted.	Statistics missing or incorrectly applied.	**10**	
	Probability & Distributions	Deep, highly accurate modeling of packet loss/arrival. Excellent simulation of the Central Limit Theorem.	Probability concepts applied clearly with minor calculation or plotting issues.	Basic application of probability models. Simulation has some flaws.	Models are poorly applied or misinterpreted. Simulation is incorrect.	Probability and distributions are skipped or completely wrong.	**15**	
	Visualizations	Visuals beautifully highlight temporal changes and class separation. Flawlessly exported as PNGs with perfect labeling.	Most visualizations are clear and appropriately labeled.	Appropriate visuals present but lack clarity or proper labeling.	Visuals are unclear, mislabeled, or poorly integrated.	Visuals are missing, irrelevant, or highly confusing.	**10**	
	Hypothesis Testing	Tests performed accurately to validate feature separation. Insightful interpretation of p-values and assumptions.	Most tests performed correctly with good interpretation.	Appropriate tests performed but interpretation is basic.	Tests applied to the wrong variables or poorly interpreted.	Hypothesis testing is missing or fundamentally flawed.	**10**	
	Regression Analysis	Logistic Regression model is robust. Deep understanding of decision boundaries and logit linearity checks.	Model constructed well. Clear findings and good interpretation.	Adequate model constructed but lacks depth in evaluating assumptions.	Regression poorly specified or interpretation of odds is incorrect.	Regression model missing or completely misapplied.	**10**	
	Report Content & Format	Exceptionally well-structured, logical flow. Flawlessly	Well-organized with clear explanations. Minor	Covers basic requirements but lacks depth. Several	Lacks structure, missing key sections, and	Highly disorganized, missing critical content.	**10**	

		adheres to formatting rules.	formatting issues.	formatting inconsistencies.	poor formatting.			
	Content Delivery (Presentation)	Engaging, professional, clear, and confident delivery. Demonstrates deep mastery of the concepts.	Clear and well-organized presentation. Good understanding.	Generally clear but lacks organization or pacing.	Unclear and disorganized. Weak understanding of the analysis.	Poor or no presentation.	**10**	
	Technical Q&A	Excellent knowledge. Easily answers complex questions regarding R code, data cleaning, and model logic.	Good knowledge. Able to answer most questions with solid reasoning.	Adequate knowledge. Struggles with deeper technical questions.	Limited knowledge. Cannot explain the rationale behind choices.	Unable to answer basic questions about the project.	**10**	
						TOTAL	**100**	